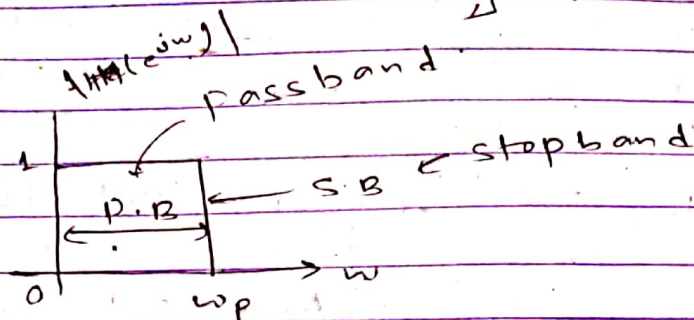
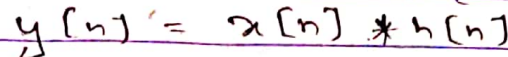


Digital Signal Analysis and Processing (DSAP): (Nabin Thapa)

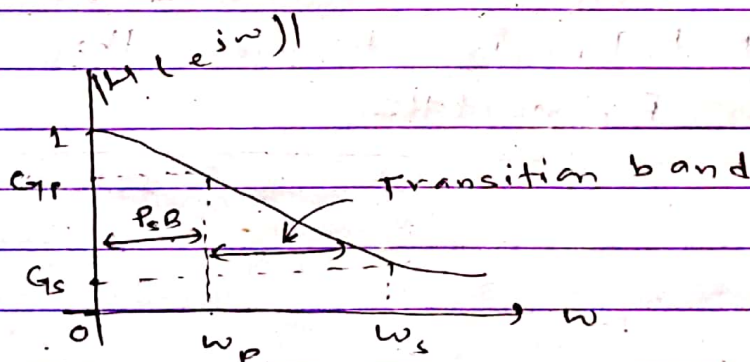
Z-transform, ~~int~~ discrete time system, IIR, FIR
DFT, ~~FFT~~

Infinite Impulse Response (IRR) Filter design

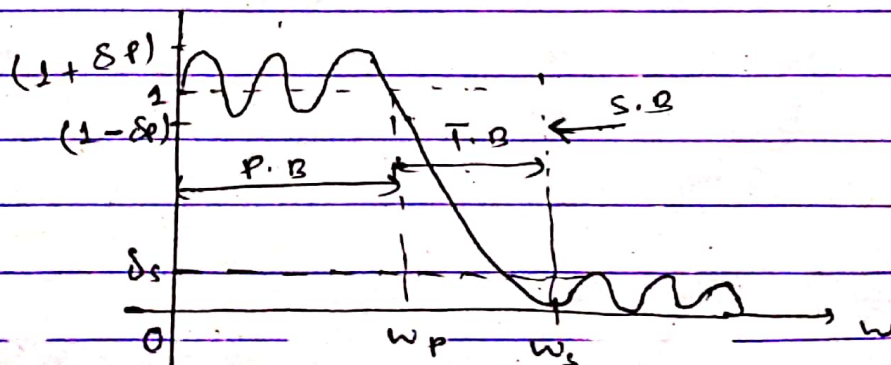


Ideal LPF (Mag. char Curve)

- $G_p = 1$ (Gain)
- $G_s = 0$
- In P.B response flat
- P.B to S.B roll-off abruptly decreases.



practical LPF (Mag. char. curve)



$\delta_p = \text{P.B ripple}$

$\delta_c = 5.8$ ripple

$\omega_p = \text{PB edge / cutoff freq. in digital domain}$

$\omega_s = \text{S.R edge / cutoff}$
 freq in digital
 domain

$\Omega_p = p \cdot B$ edge freq in analog domain
 $\Omega_s = s \cdot B$ " " " "

A_p = P.B Attenuation
 A_s = S.B Attenuation

P.B $\Rightarrow$ Pass Band
S.B $\Rightarrow$ Stop Band

In dB scale, Gain of Filter is always (-ve)

Gain = - Attenuation | in dB scale.

G_p = Gain in P.B

G_s = " " S.B

$$A_p = -20 \log \frac{1-\delta_p}{1+\delta_p} = -20 \log (1-\delta_p) + 20 \log (1+\delta_p)$$

$$A_p = -20 \log (1-\delta_p) \quad \leftarrow 1$$

$$A_s = -20 \log \delta_s \text{ dB}$$

After normalization

$T/F_{SF} \rightarrow$ sampling period / sampling frequency.

If in case, if T/F_{SF} is not given then suppose
 T as 1 sec or F_{SF} as 1 Hz.

$H_a(s)$ $H(z)$

Analog domain to digital domain transformation:

Analog domain $H_a(s)$ to digital domain $H(z)$ This transformation can be done in three different ways:

- (1) ~~Interse~~ Impulse Invariant method (IIM)
- (2) Bilinear Transformation Method (BLT)
3. Matched Z-Transform Method (MZT)

IIM method:

Formula 1:

$$H_a(s) = \frac{1}{s - p_i} \xrightarrow[\text{of Poles}]{\text{Mapping}} H(z) = \frac{1}{1 - e^{p_i T} z^{-1}}$$

Formula 2:

$$H_a(s) = \frac{s + a}{(s + a)^2 + b^2} \xrightarrow{\text{M.O.P}} H(z) = \frac{1 - e^{-aT} \cos bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$$

Formula 3:

$$H_a(s) = \frac{b}{(s + a)^2 + b^2} \xrightarrow{\text{M.O.P}} H(z) = \frac{e^{-aT} \sin bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$$

BLT-method:

$$H(z) = H_a(s)$$

$$s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$$

MZT method:

$$H_a(s) = \frac{\prod_{k=1}^M (s - z_k)}{\prod_{k=1}^N (s - p_k)} \xrightarrow{\text{M.o.p.}} H(z) = \frac{\prod_{k=1}^M (1 - e^{z_k T} z^{-1})}{\prod_{k=1}^N (1 - e^{p_k T} z^{-1})}$$

Given an analog transfer function as follows:

$$1) H_a(s) = \frac{1}{(s+1)(s+2)} = \frac{1}{s^2 + 3s + 2}$$

obtain $H(z)$.

BLT method:

$$H(z) = \frac{1}{\left(\frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} + 1 \right) \left(\frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} + 2 \right)}$$

Assume $T = 1 \text{ sec.}$

$$\therefore H(z) = \frac{1}{\left(\frac{2(1-z^{-1})}{1+z^{-1}} + 1 \right) \left(\frac{2(1-z^{-1})}{1+z^{-1}} + 2 \right)}$$

MZT method:

$$H(z) = \frac{1}{(1 - e^{-1T} z^{-1})(1 - e^{-2T} z^{-1})}$$

$$= \frac{1}{(1 - e^{-1} z^{-1})(1 - e^{-2} z^{-1})}$$

$$= \frac{1}{(1 + \alpha z^{-1})(1 - \gamma z^{-1})}$$

i.e. $\alpha = e^{-1}$, $\gamma = e^{-2} \Rightarrow$ write the values.

IMM method:

$$H_a(s) = \frac{1}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$A = -1, B = -1$$

$$\therefore H_a(s) = \frac{-1}{s+1} - \frac{1}{s+2}$$

$$= \frac{-1}{s - (-1)} - \frac{1}{s - (-2)}$$

$\Downarrow$ IMM

$$H(z) = \frac{-1}{1 - e^{-1T} z^{-1}} - \frac{1}{1 - e^{-2T} z^{-1}}$$

Assume $T = 1$

$$= \frac{-1}{1 - e^{-1} z^{-1}} - \frac{1}{1 - e^{-2} z^{-1}}$$

$$= \frac{-1}{1 - \alpha z^{-1}} - \frac{1}{1 - \gamma z^{-1}}$$

$$(2). \quad H_a(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

Soln $\rightarrow$

$$A = 0.5, \quad B = -0.5, \quad C = 0.$$

$$\begin{aligned} H_a(s) &= \frac{0.5}{s+0.5} - \frac{0.5s}{s^2+0.5s+2} \\ &= \frac{0.5}{s+0.5} - \frac{0.5s}{s^2+2 \cdot 0.5s + (0.25)^2 + (0.25)^2 + 2} \end{aligned}$$

$$\Rightarrow \frac{0.5}{s+0.5} - \frac{0.5s}{(s+0.25)^2 + (1.39)^2}$$

$$\Rightarrow \frac{0.5}{s+0.5} - 0.5 \left(\frac{s+0.25-0.25}{(s+0.25)^2 + (1.39)^2} \right)$$

$$\Rightarrow \frac{0.5}{s+0.5} - \frac{0.5s+0.25}{(s+0.25)^2 + (1.39)^2} + \frac{0.5 \times 0.25}{(s+0.25)^2 + (1.39)^2}$$

$$\Rightarrow \frac{0.5}{s+0.5} + \frac{0.5 \times 0.25}{1.39} \times \frac{1.39}{(s+0.25)^2 + (1.39)^2}$$

$$\begin{aligned} H(z) &= \frac{0.5}{1 - e^{-0.5T} z^{-1}} - 0.5 \cdot \frac{1 - e^{-0.25T} \cos(1.39T) z^{-1}}{1 - 2e^{-0.25T} \cos(1.39T) z^{-1} + e^{-2 \times 0.25T} z^{-2}} \\ &\quad + y \cdot \frac{e^{-0.25T} \sin(1.39T) z^{-1}}{1 - 2e^{-0.25T} \cos(1.39T) z^{-1} + e^{-2 \times 0.25T} z^{-2}} \end{aligned}$$

Assume $T = 1 \text{ sec.}$

$$H(z) =$$

MZT.

$$H_a(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

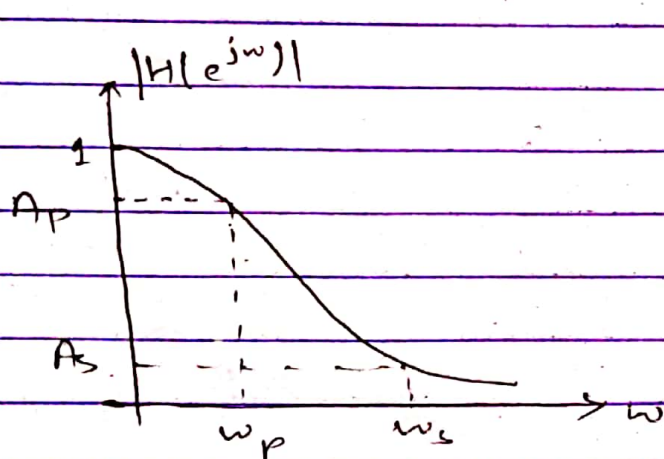
$$\Rightarrow \frac{1}{1.39(s+0.5)} \times \frac{1.39}{(s+0.25)^2 + (1.39)^2}$$

$$H(z) = \frac{1}{1.39} \times \frac{1}{1 - e^{-0.5T} z^{-1}} \times \frac{e^{-0.25T} \sin(1.39T) z^{-1}}{1 - 2e^{-0.25T} \cos(1.39T) z^{-1} + e^{-2 \times 0.25T} z^{-2}}$$

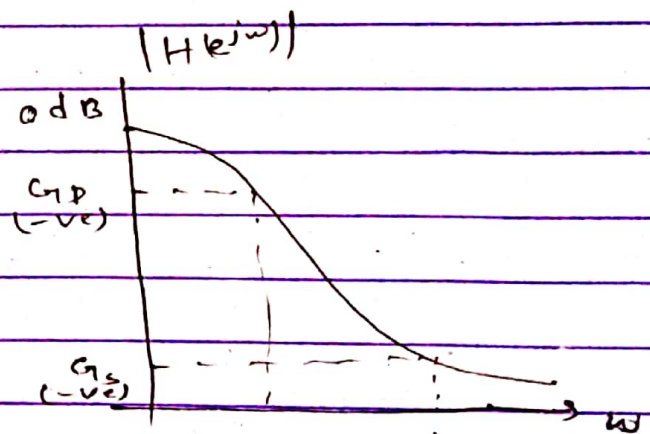
Different Approximation techniques of Filter Design

Since, the ideal characteristics of a low pass filter can't be achieved. In order to achieve practical characteristics different approximation techniques for filter design are used.

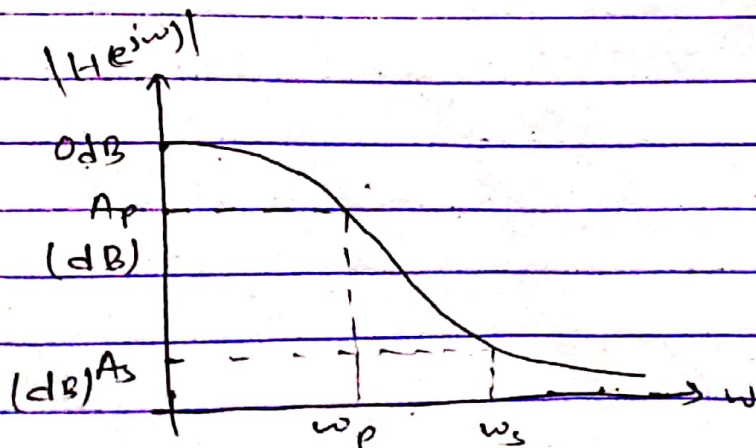
- (1) Butterworth approximation technique
- (2) Chebyshev approximation technique.
 - (a) Type I Chebyshev
 - (b) Type II Chebyshev
- (3) Elliptic / Cauer approximation technique
- (4) Bessel Thompson approximation technique



Butterworth LPF
Mag. Curve
(not in dB)



Butterworth LPF
Mag. Curve
(Gain)
(dB)



Butterworth LPF
Mag. curve (Attenuation)
(dB)

Steps to design butterworth L.P.F

Step 1: Calculate the analog domain frequencies (Ω_p and Ω_s) using digital domain frequencies (ω_p and ω_s).

IIM	BLT
$\Omega = \frac{\omega}{T}$	$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$ (Freq. Warping)
$\Rightarrow \Omega_p = \frac{\omega_p}{T}$	$\Rightarrow \Omega_p = \frac{2}{T} \tan \left(\frac{\omega_p}{2} \right)$
$\Rightarrow \Omega_s = \frac{\omega_s}{T}$	$\Rightarrow \Omega_s = \frac{2}{T} \tan \left(\frac{\omega_s}{2} \right)$

Step 2: Calculate the order of the filter (N)

Case I: If the Specifications (A_p & A_s) are in dB.

$$N = \frac{\log \left(\frac{10^{0.1 A_s} - 1}{10^{0.1 A_p} - 1} \right)}{2 \log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

Case II: If the Specifications are not in dB.

$$N = \frac{\log \left(\frac{\frac{1}{A_s^2} - 1}{\frac{1}{A_p^2} - 1} \right)}{2 \log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

~~Step~~ Step 3: Calculate the cutoff frequency (Ω_c)

Case I: If ω_c is given

IIM	BLT
$\Omega_c = \frac{\omega_c}{T}$	$\Omega_c = \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right)$

Case II: If ω_c is not given:

Case (a). If specifications are in dB.

~~BLT~~

$$\Omega_c = \Omega_p \left(\frac{10^{0.1 A_p - 1}}{1} \right)^{\frac{1}{2N}}$$

Case b: If specifications are not in dB.

$$\Omega_c = \Omega_p \left(\frac{1}{A_p^2} - 1 \right)^{\frac{1}{2N}}$$

Step 4: Calculate the pole-pair position (P_i)

$$P_i = \pm \Omega_c e^{j(N+2i+1)\frac{\pi}{2N}}$$

where, $i = 0, 1, 2, 3, \dots, (N-1)$

Step 5: Evaluate the transfer function in analog domain $H_a(s)$

$$H_a(s) = \frac{\Omega_c^N}{(s - P_0)(s - P_1)(s - P_2) \dots}$$

If Poles are Complex Conjugate pairs then

$$H_a(s) = \frac{\Omega_c^N}{(s-s_1)(s-s_1^*)(s-s_2)(s-s_2^*)\dots}$$

$s_1^* \rightarrow \text{c.c of } s_1$

$s_2^* \rightarrow \text{c.c of } s_2$

Step 6: Finally, evaluate the transfer function in digital domain $H(z)$ by using either ~~or~~ IIM or BLT method.

IIM method.

Formula 1

$$H_a(s) = \frac{1}{s-p_i} \xrightarrow{\text{M.O.P}} H(z) = \frac{1}{1-e^{p_i T} z^{-1}}$$

Formula 2:

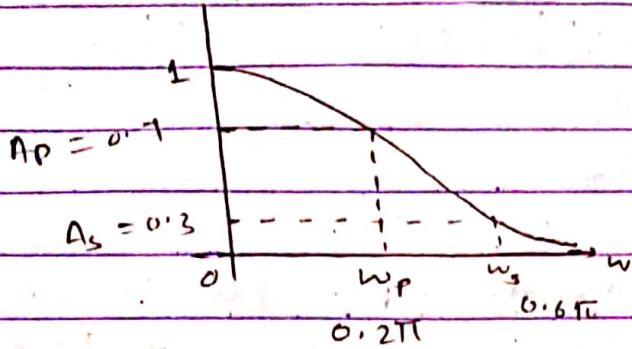
$$H_a(s) = \frac{s+a}{(s+a)^2 + b^2} \xrightarrow{\text{M.O.P}} H(z) = \frac{1-e^{-aT} \cos bT z^{-1}}{1-2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$$

Tutorial

Q.No. 14

Soln $\rightarrow$

$|H(e^{j\omega})|$



$$A_p = 0.7, \quad \omega_p = 0.2\pi$$

$$A_s = 0.3, \quad \omega_s = 0.6\pi$$

Step 1: Calculate Ω_p & Ω_s .

$$\Omega_p = \frac{\omega_p}{T} = \frac{0.2\pi}{1} = 0.2\pi$$

$$\Omega_s = \frac{\omega_s}{T} = \frac{0.6\pi}{1} = 0.6\pi$$

Step 2: Calculate order N .

$$N = \frac{\log \left(\frac{1 - 1}{A_s^2} \right)}{2 \log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$\approx 1.034 \approx 2$$

Step 3: Calculate Ω_c .

$$\Omega_c = \frac{\Omega_p}{\left(\frac{1}{\Omega_p^2} - 1 \right)^{\frac{1}{2N}}}$$

$$\Omega_c = 0.622$$

Step 4. Calculate Pole-Pair position.

$$P_i = \pm \Omega_c e^{j(N+2i-1)\frac{\pi}{2N}}$$

$$i = 0, 1 \quad \text{as } N = 2.$$

$$= \pm 0.622 e^{j(3+2i)\frac{\pi}{4}}$$

For $i = 0$,

$$P_0 = \pm 0.622 e^{j\frac{3\pi}{4}}$$

$$= \pm 0.622 \left(\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4} \right)$$

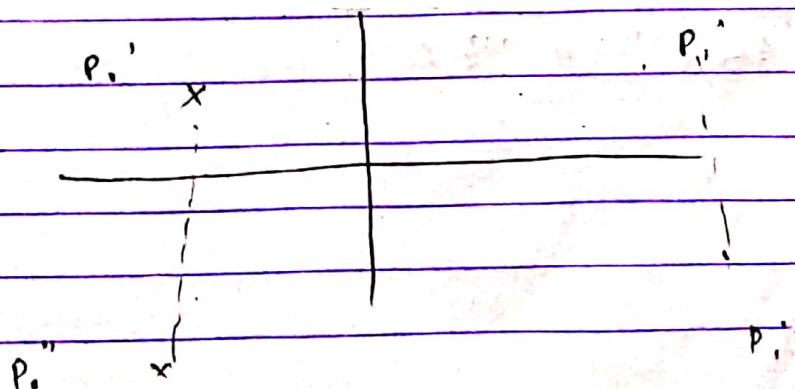
$$= \pm (-0.439 + j0.439)$$

$$= \begin{cases} -0.439 + j0.439 & P_0' \\ -0.439 - j0.439 & P_0'' \end{cases}$$

For $i = 1$

$$P_i = \pm 0.622 e^{j\frac{5\pi}{4}}$$

$$= \begin{cases} -0.439 - j0.439 & P_1' \\ 0.439 + j0.439 & P_1'' \end{cases}$$



Pole-pair plot in s-domain.

~~Step 5~~: Note: In order to design stable IIR filter, poles of left half of s-plane are only considered.

Step 5:

$$\begin{aligned}
 H_a(s) &= \frac{K_c}{(s - (-0.439 + j0.439))(s - (-0.439 - j0.439))} \\
 &= \frac{(0.622)^2}{(s + 0.439 - j0.439)(s + 0.439 + j0.439)} \\
 &= \frac{(0.622)^2}{[(s + 0.439)^2 + (0.439)^2]}
 \end{aligned}$$

Step 6:

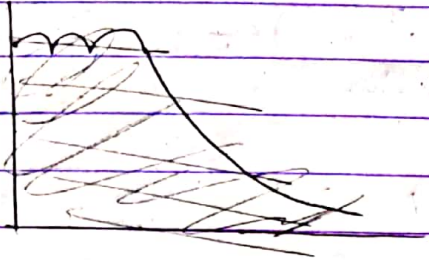
$$H_a(s) = \frac{(0.622)^2}{0.439} \times \frac{0.439}{(s + 0.439)^2 + 0.439^2}$$

↓ IM

$$H(z) = \frac{(0.622)^2}{0.439} \times \frac{e^{-0.439} \sin(0.439) z^{-1}}{1 - 2e^{-0.439} \cos(0.439) z^{-1} + e^{-2 \times 0.439} z^{-2}}$$

$$\begin{aligned}
 H(z) &\Rightarrow 0.241 z^{-1} \\
 &1 - 1.165 z^{-1} + 0.418 z^{-2}
 \end{aligned}$$

(9) Solⁿ → Using BLT method.



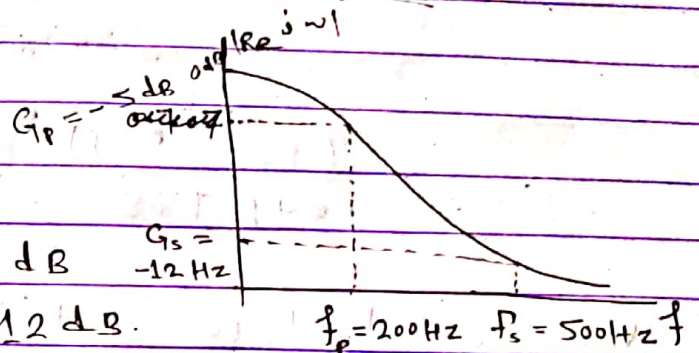
Given that,

$$F_{sp} = 5 \text{ kHz}$$

$$G_p = -5 \text{ dB} \Rightarrow A_p = 5 \text{ dB}$$

$$G_s = -12 \text{ dB} \Rightarrow A_s = 12 \text{ dB}$$

$$\omega_p = 2\pi f_p \approx 2\pi \frac{f_p}{F_{sp}} \Rightarrow 0.08\pi$$



$$\omega_s = 2\pi f_s \Rightarrow 2\pi \frac{f_s}{F_{sp}} \Rightarrow 0.2\pi$$

Step 1: Calculate Ω_p & Ω_s

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right)$$

$$= 2 \times 5 \times 10^3 \tan\left(\frac{0.08\pi}{2}\right)$$

$$= 1263.29$$

$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) \Rightarrow 3249.19$$

Step 2: Calculate order(N)

$$N = \frac{\log \left(\frac{10^{0.1 A_s} - 1}{\log 10^{0.1 A_p} - 1} \right)}{2 \log \left(\frac{\Omega_s}{\Omega_p} \right)} = 1.019 \approx 2$$

~~Step~~ step 3: Calculate Ω_c .

$$\begin{aligned} \Omega_c &= \frac{\Omega_p}{(10^{0.1 A_p} - 1)^{\frac{1}{2N}}} \\ &= 1041.778 \end{aligned}$$

Step 4: Calculate Pole-Pair position (p_i)

$$\begin{aligned} p_i &= \pm \Omega_c e^{j(N+2i)\frac{\pi}{2N}} \\ &= \pm 1041.778 e^{j(3+2i)\frac{\pi}{4}} \end{aligned}$$

$i = 0, 1$

~~Now~~ Now, for $i = 0$

$$\begin{aligned} p_i &= \pm 1041.778 e^{j(3\pi/4)} \\ &= \pm 1041.778 \left(\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4} \right) \\ &= \pm (-736.65 + j 736.65) \end{aligned}$$

$$= \begin{cases} -736.65 + j 736.65 & p_0^* \\ 736.65 - j 736.65 & p_0^{**} \end{cases}$$

Similarly

$$p_i = \begin{cases} -736.65 - j 736.65 & p_1^* \\ 736.65 + j 736.65 & p_1^{**} \end{cases}$$

In order to design a stable IIR filter poles of left half of s-plane are only considered



pole-pair plot in s-domain.

step 5 : Calculate $H_A(s)$.

$$H_A(s) = \frac{(1041.78)^2}{(s + 736.65)^2 + (736.65)^2}$$

step 6 : Calculate $H(z)$.

$$H(z) = \frac{(1041.78)^2}{\left(\frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} + 736.65 \right)^2 + (736.65)^2}$$

$$H(z) = \frac{(1041.78)^2}{\left(2 \times 5 \times 10^3 \frac{1-z^{-1}}{1+z^{-1}} + 736.65 \right)^2 + (736.65)^2}$$

simplify the eqⁿ.

=

Digital Domain Transformation:

S.No	Types of Filter	Transformation	Parameters.
1.	LPF	$z^{-1} \rightarrow \frac{z^{-1} - a}{1 - az^{-1}}$	$a = \frac{\sin \frac{\omega_p - \omega_p'}{2}}{\sin \frac{\omega_p + \omega_p'}{2}}$
2.	HPF	$z^{-1} \rightarrow \frac{-z^{-1} - a}{1 + az^{-1}}$	$a = \frac{-\cos \frac{\omega_p + \omega_p'}{2}}{\cos \frac{\omega_p - \omega_p'}{2}}$
3.	BPF	$z^{-1} \rightarrow \frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$\alpha = \frac{\cos \frac{\omega_u + \omega_L}{2}}{\cos \frac{\omega_u - \omega_L}{2}};$ $k = \cot \left(\frac{\omega_u - \omega_L}{2} \right) \cdot \tan \left(\frac{\omega_p}{2} \right)$ $a_1 = \frac{2\alpha F}{k+1}, a_2 = \frac{F-1}{k+1}$
4.	BSF	$z^{-1} \rightarrow \frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$\alpha = \text{Same as BPF},$ $k = \tan \left(\frac{\omega_u - \omega_L}{2} \right) \cot \left(\frac{\omega_p}{2} \right)$ $a_1 = \frac{2\alpha k}{k+1}, a_2 = \frac{1-k}{1+k}$

Q.No-14, 2nd part

$$\omega_p = 0.2\pi$$

$$\omega_p' = 0.45\pi$$

$$H(z) = \frac{0.24 z^{-1}}{1 - 1.165 z^{-1} + 0.41 z^{-2}}$$

$$1 - 1.165 z^{-1} + 0.41 z^{-2}$$

New, LPF,

$$H'(z) = \frac{0.24 \frac{z^{-1} - a}{1 - az^{-1}}}{1 - 1.165 \frac{z^{-1} - a}{1 - az^{-1}} + 0.41 \left(\frac{z^{-1} - a}{1 - az^{-1}} \right)^2}$$

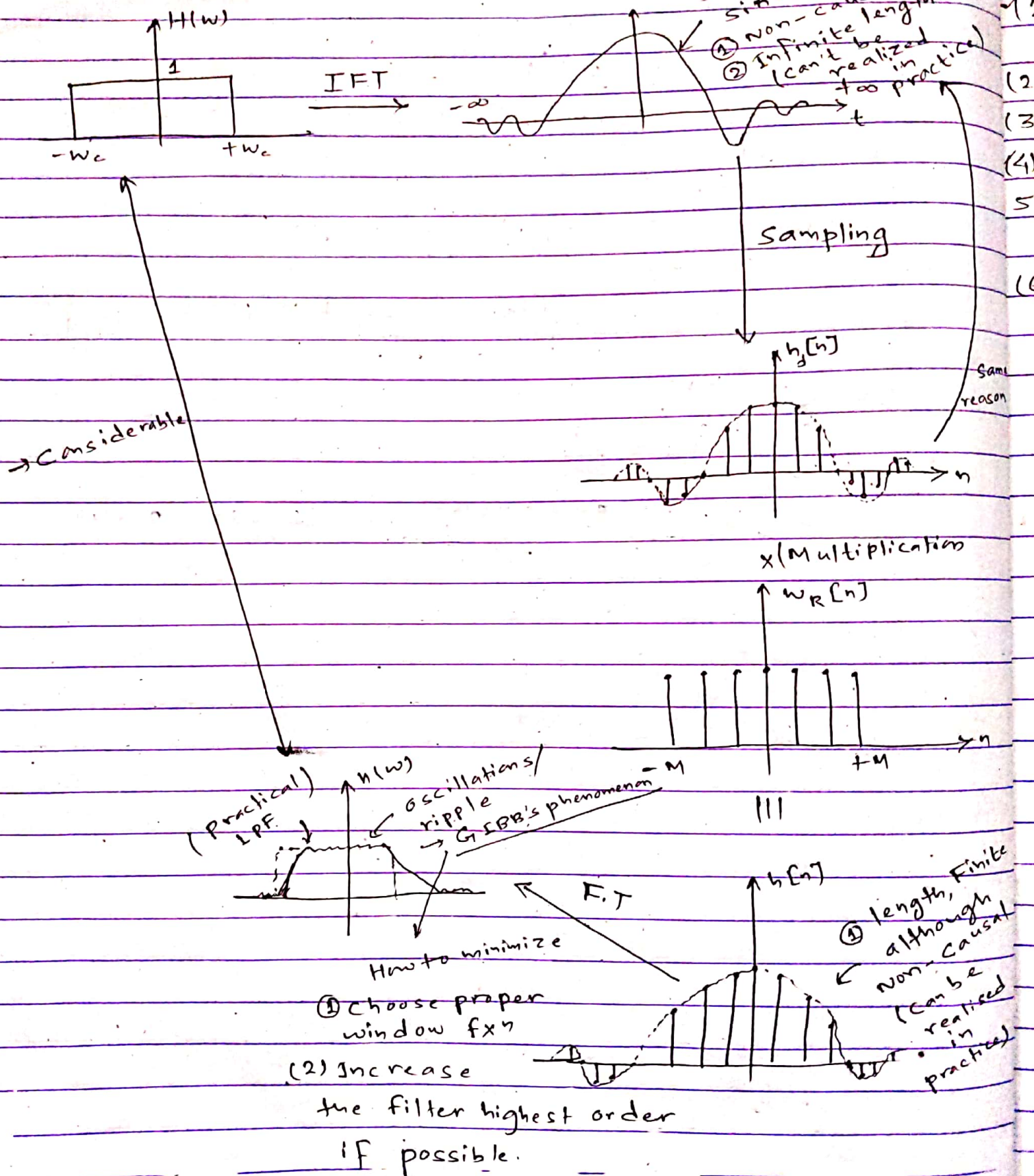
$$1 - 1.165 \frac{z^{-1} - a}{1 - az^{-1}} + 0.41 \left(\frac{z^{-1} - a}{1 - az^{-1}} \right)^2$$

$$a = \frac{\sin \frac{\omega_p - \omega_p'}{2}}{\sin \frac{\omega_p + \omega_p'}{2}}$$

$$\Rightarrow \frac{\sin \frac{0.2\pi - 0.45\pi}{2}}{\sin \frac{0.2\pi + 0.45\pi}{2}} \Rightarrow \cancel{0.448} - 0.448$$

Chapter 5

FIR Filter design



✓(1) Impulse response of Low pass filter is non-causal / sinc-pulse?

(2) GIBB's phenomenon.

(3) Desired Impulse Response of LPF, HPF, BPF, BSF.

(4) Condition for Linear phase FIR Filter.

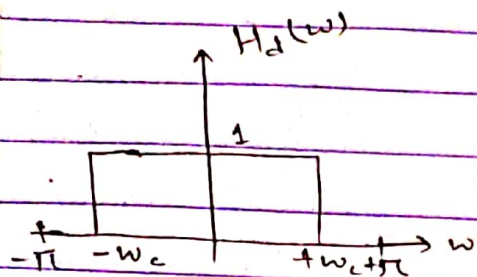
5) FIR Filter design using (a) windowing technique.

(b) Frequency sampling method.

(6) Remez exchange algorithm / optimal method
algorithm / parks ~~and~~ McClellan's algorithm

Desired impulse response of different type of ideal filter:

~~I~~ (1) LPF:



$$H_d(w) = \begin{cases} 1, & |w| \leq w_c \Rightarrow -w_c \leq w \leq w_c \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} h_d[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(w) e^{jwn} dw \\ &= \frac{1}{2\pi} \left[\int_{-\pi}^{-w_c} 0 \times e^{jwn} dw + \int_{-w_c}^{w_c} 1 \times e^{jwn} dw + \int_{w_c}^{\pi} 0 \times e^{jwn} dw \right] \\ &= \frac{1}{2\pi} \int_{-w_c}^{w_c} e^{jwn} dw \end{aligned}$$

$$= \frac{1}{2\pi} \int_{-w_c}^{w_c} (\cos wn + j \sin wn) dw$$

$$= \frac{1}{\pi n} [\sin w_c n]$$

$$= \frac{1}{\pi n} \times \frac{\sin(w_c n)}{w_c n} \times w_c n$$

$$= \frac{w_c}{\pi} \times [\text{sinc}(w_c n)]$$

$$= \frac{2\pi f_c}{\pi} [\text{sinc}(w_c n)] \Rightarrow h_d[n] = 2f_c [\text{sinc}(w_c n)]$$

For $n \neq 0$.

For $n = 0$

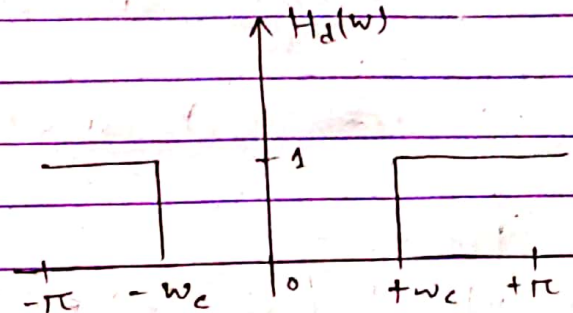
$$h_d[n] = 2F_c [\overset{1}{\cancel{\sin c 0}}]$$

$$h_d[n] = 2F_c, n = 0$$

(2) HPF

For $n \neq 0$

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$



$$= \frac{1}{2\pi} \left[\int_{-\pi}^{-wc} H_d(\omega) e^{j\omega n} d\omega + \int_{-wc}^0 H_d(\omega) e^{j\omega n} d\omega + \int_0^{wc} H_d(\omega) e^{j\omega n} d\omega + \int_{wc}^{\pi} H_d(\omega) e^{j\omega n} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{-wc} H_d(\omega) e^{j\omega n} d\omega + \int_{wc}^{\pi} H_d(\omega) e^{j\omega n} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{-wc} e^{j\omega n} d\omega + \int_{wc}^{\pi} e^{j\omega n} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{-wc} (\cos \omega n + j \sin \omega n) d\omega + \int_{wc}^{\pi} (\cos \omega n + j \sin \omega n) d\omega \right]$$

$$= \frac{1}{2\pi} \left[\left(\frac{e^{-j\omega n}}{jn} - \frac{e^{-j\omega n}}{-jn} \right) + \left(\frac{e^{j\omega n}}{jn} - \frac{e^{j\omega n}}{-jn} \right) \right]$$

$$= \frac{1}{2\pi \cdot jn} \left[e^{-j\omega n} - e^{j\omega n} + e^{j\omega n} - e^{-j\omega n} \right]$$

$$= \frac{1}{2\pi \cdot jn} \left[\frac{e^{j\pi n} - e^{-j\pi n}}{2j} + \frac{e^{-j\omega n} - e^{j\omega n}}{2j} \right]$$

$$= \frac{1}{\pi n} (\sin n\pi - \sin \omega_c n)$$

$$\Rightarrow \frac{1}{\pi n} \sin \omega_c n$$

$$= \frac{-1}{\pi n} \frac{\sin \omega_c n}{\omega_c n} \times \omega_c n$$

$$= -\frac{2\pi f_c}{\pi} \text{sinc}(\omega_c n)$$

$$= -\cancel{2\pi} - 2F_c \text{sinc}(\omega_c n)$$

For $n = 0$;

$$h_d(n) = \frac{1}{2\pi} \left[\int_{-\pi}^{-\omega_c} 1 \times e^{j\omega \times 0} d\omega + 0 + \int_{\omega_c}^{\pi} 1 \times e^{j\omega \times 0} d\omega \right]$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{-\omega_c} d\omega + \int_{\omega_c}^{\pi} d\omega \right]$$

$$= \frac{1}{2\pi} \left[(-\omega_c + \pi) + (\pi - \omega_c) \right]$$

$$= \frac{1}{2\pi} (\pi - \omega_c) \Rightarrow \frac{1}{\pi} (\pi - 2\pi f_c)$$

$$\Rightarrow 1 - 2F_c$$

Summary:

S.No.	Types of Filter	$h_d(n)$ $n \neq 0$	$h_d(n)$ $n = 0$
1.	LPF	$2F_c \text{sinc}(\omega_c n)$	$2F_c$
2.	HPF	$-2F_c [\text{sinc}(\omega_c n)]$	$1 - 2F_c$
3.	BPF	$2F_{c2} [\text{sinc}(\omega_{c2} n)]$ $- 2F_{c1} [\text{sinc}(\omega_{c1} n)]$	$2(F_{c2} - F_{c1})$
4.	BSF	$2F_{c1} [\text{sinc}(\omega_{c1} n)] -$ $2F_{c2} [\text{sinc}(\omega_{c2} n)]$	$1 + 2(F_{c2} - F_{c1})$

Table 1: Desired Impulse Response of Ideal filters.

S.N.	Window's type	Windows function $-\frac{M-1}{2} \leq n \leq \frac{M-1}{2}$	Window length (M)	Stopband Attenuation	Main Lobe width ($\omega_s - \omega_p$)
1)	Rectangular	$w_R[n] = 1$	$M = \frac{0.9}{\Delta f}$	21 dB	$\frac{4\pi}{M}$
2)	Hanning	$w_{HAN}[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right)$	$M = \frac{3.1}{\Delta f}$	44 dB	$\frac{8\pi}{M}$
3)	Hamming	$w_{HAM}[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right)$	$M = \frac{3.3}{\Delta f}$	53 dB	$\frac{8\pi}{M}$
4)	Blackmann	$w_B[n] = 0.42 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right) + 0.08 \cos\left(\frac{4\pi n}{M-1}\right)$	$M = \frac{5.5}{\Delta f}$	74 dB	$\frac{12\pi}{M}$
5)	Kaiser	$w_K[n] = \frac{I_0\left[\beta \sqrt{1 - \left(\frac{2n}{M-1}\right)^2}\right]}{I_0(\beta)}$	DIFFERENT APPROACH		

Table 2.

$\Delta f = f_s - f_p$ (LPF)	$\Delta \omega = \omega_s - \omega_p$
$\Delta f = f_p - f_s$ (HPF)	$\Delta \omega = \omega_p - \omega_s$

$$h[n] = h_d[n] \times w[n]$$

Tut.

Q 1.

Step 1: The problem gives desired frequency response ($H_d(e^{j\omega})$), types of window and the order of the filter (M).

The window function are defined for the range,
for odd order (M)

$$\text{Range: } -\frac{M-1}{2} \leq n \leq \frac{M-1}{2}$$

eg: If $M = 7$; Range : $-3 \leq n \leq 3$

for even order (M)

$$\text{Range: } -\left(\frac{M}{2} - 1\right) \leq n \leq \frac{M}{2}$$

OR

$$-\frac{M}{2} \leq n \leq \left(\frac{M}{2} - 1\right)$$

eg: If $M = 6$

$$-2 \leq n \leq 3 \quad \text{or} \quad -3 \leq n \leq 2$$

Step 2: Calculate the desired impulse response of the respective filter (LPF, HPF, BPF, BSF) using the formula;

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega.$$

From: Fig: Table 1

Step 3: Select the proper window function by observing the value of stopband Attenuation (A)

Fig: Table 2. (except KASER window)

Step 4: Finally the impulse response of the FIR filter is obtained by using the expression

$$h[n] = h_d[n] \times w[n]$$

FIR filter design using KASER window method.

Step 1: Determine the minimum value of the ripple (δ_p, δ_s)

$$\delta_s = 10^{-0.05 A_s}, \quad \delta_p = \frac{10^{0.05 A_p} - 1}{10^{0.05 A_p} + 1}$$

$$A_s = -20 \log \delta_s, \quad A_p = 20 \log \left(\frac{1 + \delta_p}{1 - \delta_p} \right)$$

$$\delta = \min(\delta_p, \delta_s)$$

Step 2: Calculate the Attenuation value in dB.

$$A = -20 \log(\delta) \text{ dB}$$

Step 3: Calculate the ~~K~~ KAISER parameter (β) from the following eqn.

$$\beta = \begin{cases} 0.110 (A - 8.7) & A > 50 \\ 0.5842 (A - 21)^{0.4} + 0.07886 (A - 21) & 21 \leq A \leq 50 \\ 0 & A < 21 \end{cases}$$

Step 4: Calculate the order of the filter using the expression

$$M = \frac{A - 8}{2.285 D\omega}$$

$$D\omega = \omega_s - \omega_p \text{ (LPF)}$$

$$D\omega = \omega_p - \omega_s \text{ (HPF)}$$

(Note: Length of the filter for KAISER window $(L) = M + 1$)

Step 5: The window function of KAISER is defined as

$$w_{KAI}[n] = \begin{cases} \frac{I_0\left(\beta \sqrt{1 - \left(\frac{2n}{M-1}\right)^2}\right)}{I_0(\beta)} & |n| \leq \frac{M-1}{2} \\ 0 & \text{elsewhere} \end{cases}$$

$$I_0(\beta) = 1 + \frac{(0.25\beta^2)^2}{1!} + \frac{(0.25\beta^2)^4}{(2!)^2} + \frac{(0.25\beta^2)^6}{(3!)^2} + \dots$$

Step 6: Evaluate the desired impulse response of the respective filter (LPF, HPF, BPF, BSF)

using the expression

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega$$

Step 7: finally the impulse response of FIR filter is

$$H[n] = H_d[n] \times w[n]$$

Tut:

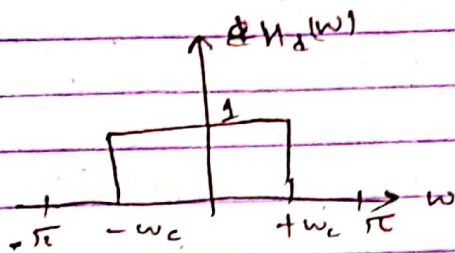
Q. No. 6.

$$F_p = 2 \times 10^3 \text{ Hz}, F_s = 5 \times 10^3, A_s = 42 \text{ dB}$$

$$F_{sf} = 20 \text{ kHz}$$

Since $F_s > F_p$, FIR LPF is designed.

Desired impulse response of LPF is $(H_d[n])$



$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(w) e^{jwn} dw$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^{-wc} 0 \times e^{jwn} dw + \int_{-wc}^{wc} 1 \times e^{jwn} dw + \int_{wc}^{\pi} 0 \times e^{jwn} dw \right]$$

$$= \frac{1}{2\pi} \int_{-wc}^{wc} e^{jwn} dw$$

$$\Rightarrow \frac{1}{2\pi} \int_{-wc}^{wc} (\cos wn + j \sin wn) dw$$

$$\frac{2\pi f_c}{\pi} (\text{sinc}(\omega_c n))$$

$$H_d(n) = \frac{2f_c}{\omega_c} (\text{sinc}(\omega_c n)) \quad \text{for } n \neq 0$$

$$H_d(n) = 2F_c, \quad \text{for } n = 0$$

$$H_d(n) = \begin{cases} \frac{2F_c \sin(\omega_c n)}{\omega_c}, & \text{for } n \neq 0 \\ 2F_c, & \text{for } n = 0 \end{cases}$$

$$F_p = \frac{f_p}{F_{SF}} = 0.1 \Rightarrow \omega_p = 2\pi F_p \Rightarrow 2\pi \times 0.1 \Rightarrow 0.2\pi$$

$$F_s = \frac{f_s}{F_{SF}} = 0.25 \Rightarrow \omega_s = 2\pi F_s \Rightarrow 2\pi \times 0.25 = 0.5\pi$$

$$F_c = \frac{F_p + F_s}{2} \Rightarrow \omega_c = 2\pi F_c \Rightarrow 2\pi \times 0.175 \Rightarrow 0.35\pi$$

$$h_d(n) = \begin{cases} \frac{0.35 \sin(0.35\pi n)}{0.35\pi n}, & n \neq 0 \\ 0.35, & n = 0 \end{cases} = \begin{cases} \frac{\sin(0.35\pi n)}{\pi n}, & n \neq 0 \\ 0.35, & n = 0 \end{cases}$$

Given that $A_s = 42 \text{ dB}$ i.e. at least 42 dB Attenuation is required in stop band it is obvious that hanning window is used because it provides minimum attenuation of 44 dB which is expressed as

$$w_{\text{hann}}(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right) \quad \text{for}$$

$$\begin{cases} -\frac{M-1}{2} \leq n \leq \frac{M-1}{2} \\ 0, \text{ else} \end{cases}$$

Where

$$M = \frac{3.1}{\Delta f} \Rightarrow \frac{3.1}{F_s - F_p} = \frac{3.1}{0.25 - 0.1} = 20.67 \approx 21$$

$$w_{HAN} = \begin{cases} 0.5 - 0.5 \cos(0.1\pi n), & -10 \leq n \leq +10 \\ 0, & \text{elsewhere} \end{cases}$$

Finally, $H[n] = H_d[n] \times w[n]$

$$= \begin{cases} \frac{\sin(0.35\pi n)}{0.25\pi n} \times [0.5 - 0.5 \cos(0.1\pi n)], & -10 \leq n \leq 10 \\ & n = 0 \end{cases}$$

Q. no. 19:

$$0.899 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.2\pi$$
$$|H(e^{j\omega})| \leq 0.01, \quad 0.4\pi \leq \omega \leq \pi$$

$$\omega_p = 0.2\pi \Rightarrow \text{LPF.}$$

$$\omega_s = 0.4\pi$$

$$h_d(n) = \begin{cases} 2F_c \frac{\sin \omega_c n}{\omega_c n}, & n \neq 0 \\ 2F_c, & n = 0 \end{cases}$$

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0.2\pi + 0.4\pi}{2} = 0.3\pi.$$

$$F_c = \frac{\omega_c}{2\pi} = \frac{0.3\pi}{2\pi} = 0.15$$

$$h_d(n) = \begin{cases} \frac{\sin(0.3\pi n)}{\pi n}, & n \neq 0 \\ 0.3, & n = 0 \end{cases}$$

$w(n)$

$$\text{Step 1: } \delta_p = 0.899, \quad \delta_s = 0.01$$
$$\delta = 0.01$$

$$\text{Step 2: } A_p = -20 \log(\delta)$$
$$= 40 \text{ dB}$$

Step 3:

$$B = 0.5842 (A - 21)^{0.4} + 0.07886 (A - 21)$$
$$= 0.5842 (40 - 21)^{0.4} + 0.07886 (40 - 21)$$
$$\Rightarrow 3.39$$

Step 4:

$$M = \frac{A - 8}{2.285 \Delta \omega} = \frac{A - 8}{2.285 (\omega_s - \omega_p)}$$

$$= 22.28 \approx 23$$

Step 5:

$$w_r(n) = \begin{cases} I_0(3.39) \left[1 - \left(\frac{n}{11} \right)^2 \right] & -11 \leq n \leq 11 \\ I_0(3.39) & 0 \end{cases}$$

finally,

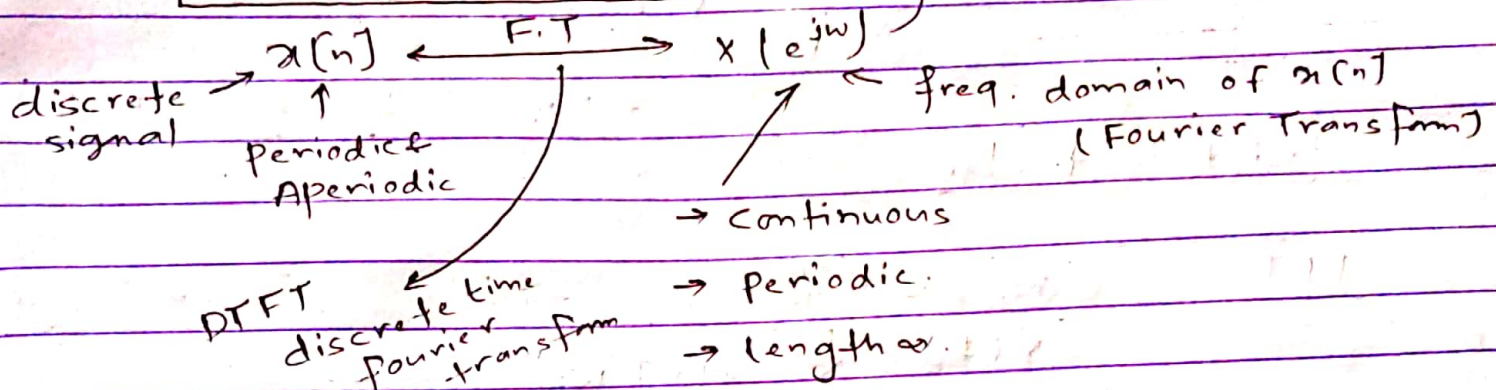
$$h(n) = h_1(n) \times w_r(n)$$

$$h(n) = \begin{cases} \frac{\sin(0.3\pi n)}{\pi n} \times \frac{I_0(3.39) \left[1 - \left(\frac{n}{11} \right)^2 \right]}{I_0(3.39)} & -11 \leq n \leq 11 \\ 0.3 & n = 0 \end{cases}$$

Why DFT although you've DTFT.

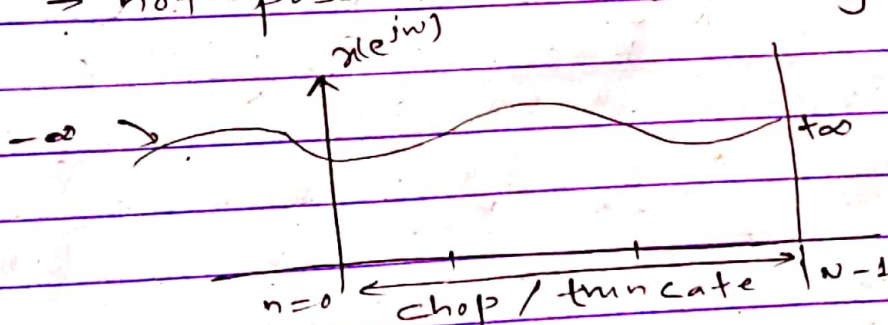
Discrete Fourier Transform

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$



Available

- digital instruments
- not possible because length ∞ .



DFT (discrete Fourier transform)

DFT

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

where $k = 0, 1, 2, \dots, N-1$

$$x[n] \xrightarrow[N\text{-point DFT}]{} X(k) \quad \text{--- (1)}$$



Inverse discrete Fourier Transform (IDFT)

$$X(k) \xrightarrow{\text{IDFT}} x(n) \text{ --- (2)}$$

where

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi}{N} nk}$$

$$n = 0, 1, 2, 3, \dots, N-1$$

Twiddle factor

DFT

$$W_N = e^{-j \frac{2\pi}{N}}$$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk} \text{ --- (3)}$$

IDFT

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-nk} \text{ --- (4)}$$

Examples:

Find DFT:

$$(1) x(n) = \{1, 1\}$$

$$(2) x(n) = u(n) - u(n-2)$$

$$(3) x(n) = \cos \frac{\pi n}{4}$$

$$(4) x(n) = \delta(n+1) - 2\delta(n) + \delta(n-2)$$

$$(5) x(n) = \{1, 2, 3\}$$

$$(6) x(n) = \{1, 2, 3, 4, 1\}$$

Adding zero on the sequence is called zero padding.

In Matrix Form:

$$X_k = [W_N] x_n$$

$x(0)$		1	1	1	1		1	$x(0)$
$x(1)$		1	w_N^1	w_N^2	w_N^3	...	w_N^{N-1}	$x(1)$
$x(2)$		1	w_N^2	w_N^4	w_N^6	...	$w_N^{2(N-1)}$	$x(2)$
$\vdots$		1	w_N^3	w_N^6	w_N^9	...	$w_N^{3(N-1)}$	$\vdots$
$x(N-1)$		1	w_N^{N-1}	$w_N^{2(N-1)}$			$w_N^{(N-1)^2}$	$x(N-1)$

(1). Soln $\rightarrow$

$$X_k = [W_N] x_n$$

~~Ans~~

$$X(k) = \begin{bmatrix} 1 & 1 \\ 1 & w_2^1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$w_N = e^{-j \frac{2\pi}{N}}$$

$$w_2 = e^{-j\pi}$$

$$w_2^1 = e^{-j\pi + 1}$$

$$= \cos \pi + j \sin \pi$$

$$x(k) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\therefore x(k) = \{2, 0\}$$

In time domain, it doesn't have any significance but in freq. domain spectrum becomes ~~time~~ fine because no. of samples in freq. domain ~~area~~ increases.

$$\Rightarrow (2) \quad x[n] = u[n] - u[n-2], \quad N=4.$$

$$x[n] = \{1, 1, 0, 0\}$$

$$X(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^1 & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$w_N = e^{-j2\pi/N}$$

$$w_4 = e^{-j\pi/2}$$

$$w_4^1 = w_4 \times w_4^0 = e^{-j\pi/2} \times 1 = \cos \frac{\pi}{2} - j \sin \frac{\pi}{2} = -j$$

$$w_4^2 = w_4^1 \times w_4^1 = -j \times -j = -1$$

$$w_4^3 = w_4^2 \times w_4^1 = -1 \times -j = j$$

$$w_4^4 = w_4^3 \times w_4^1 = j \times -j = 1$$

$$w_4^6 = w_4^4 \times w_4^2 = 1 \times -1 = -1$$

$$w_4^9 = (w_4^3)^3 = (j)^3 = -j$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & +j \\ 1 & -1 & 1 & -1 \\ 1 & +j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 \\ 1-j \\ 1+j \end{bmatrix}$$

$$x(k) = \{2, 1-j, 1+j\}$$

$$(4) \quad x[n] = \delta[n+1] - 2\delta[n] + \delta[n-2]$$

$$n = -1, \quad x[-1] = \delta[0] - 2\delta[-1] + \delta[-3] \\ = 1 - 0 + 0 = 1$$

$$n = 0, \quad x[0] = \delta[1] - 2\delta[0] + \delta[-2] \\ = 0 - 2 + 0 = -2$$

$$n = 1, \quad x[1] = \delta[2] - 2\delta[1] + \delta[-1] \\ = 0 - 0 + 0 = 0$$

$$n = 2, \quad x[2] = \delta[3] - 2\delta[2] + \delta[0] \\ = 0 - 0 + 1 = 1$$

$$\therefore x[n] = \{1, -2, 0, 1\}$$

4. Calculate DFT.

IDFT:

$$x_N = \frac{1}{N} [w_N^*] X_k$$

↑ conjugate matrix of (w_N)

$$x[k] = \{2, 1-j, 0, 1+j\} \quad N=4$$

$$x[n] = \frac{1}{N} [w_N^*] X_k$$

$$[w_4] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & j \end{bmatrix}$$

$$[w_4^*] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & +j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & +j \end{bmatrix}$$

$$x[n] = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 2 \\ 1-j \\ 0 \\ 1+j \end{bmatrix} \\ = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

$$L = m + n - 1$$

$$x[n] = \{1, 2, 3\}$$

$$h[n] = \{1, 2\}$$

$x[n] \backslash h[n]$	1	2
1	1	2
2	2	4
3	3	6

$$y[n] = \{1, 4, 7, 6\}$$

Linear Convolution:

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

Range of n :

$$n_L = -1, n_R = 1 \quad (\text{see the position from origin as left is taken})$$

$$y_L = x_L + h_L = -1 + (-1) = -2$$

$$y_R = x_R + h_R = 1 + 0 = 1$$

$$(-2 \text{ to } 1)$$

Range of k : $(-1 \text{ to } 1)$

$$y[n] = \sum_{k=-1}^1 x[k] h[n-k]$$

$$n = -1$$

$$y[-1] = \sum_{k=-1}^1 x[k] h[-1-k]$$

$$= x[-1] h[-2] + x[0] h[-1] + x[1] h[0]$$

$$= 1 \times 2 + 2 \times 1 + 3 \times 0$$

$$= 2 + 2 = 4$$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k] = h[n] * x[n]$$

Soln $\rightarrow$

Range of n : $(-2 \text{ to } 1)$

Range of k : $(-1 \text{ to } 0)$

$$n = -1$$

$$\begin{aligned} y[-1] &= \sum_{k=-1}^0 h[k] x[-1-k] \\ &= h[-1] x[0] + h[0] x[-1] \\ &= 1 \times 2 + 2 \times 1 \\ &= 4 \end{aligned}$$

$$(1) \quad x[n] = \{1, 2, 3\}$$

$$h[n] = \{4, 5\} \Rightarrow \{4, 5, 0\}$$

$$(2) \quad x[n] = \{1, 2, 3\}$$

$$h[n] = \{4, 5\} \Rightarrow \{4, 5, 0\}$$

$$(3) \quad x[n] = \{1, 2, 3, 4\}$$

$$h[n] = \{4, 5, 6, 7\}$$

$$(4) \quad x[n] = \{1, 2, 3, 4, 5\}$$

$$h[n] = \{1, 2, 3, 4\}$$

$$(5) \quad x[n] = \{1, 2, 3, 4\}$$

$$h[n] = \{1, 2, 3\}$$

$$(6) \quad x[n] = \{1, 2, 3, 4\}$$

$$h[n] = \{5, 6, 7, 8\}$$

$h(-n) \Rightarrow$ horizontal same, vertical change.

Circular Convolution:

$$y[m] = x[n] \otimes h[n]$$

$$= \sum_{n=0}^{N-1} x[n] h[(m-n)_N]$$

① $x[n] = \{1, 2\}$ & $h[n] = \{3, 4\}$

$m = 0, 1.$

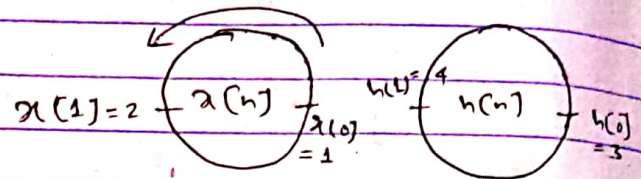
for, $m = 0$

$$y[0] = \sum_{n=0}^{N-1} x[n] h[-n]$$

$$y[0] = 1 \times 3 + 2 \times 4$$

$$= 3 + 8$$

$$= 11$$



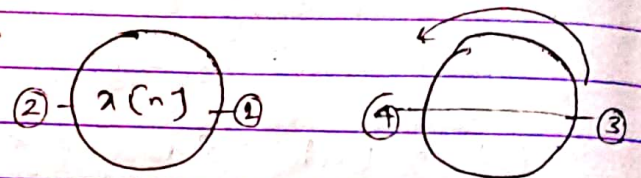
for $m = 1$

$$y[1] = \sum_{n=0}^{N-1} x[n] h[1-n]$$

$$y[1] = 1 \times 4 + 2 \times 3$$

$$= 4 + 6$$

$$= 10$$

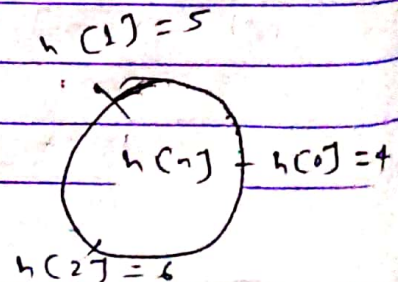
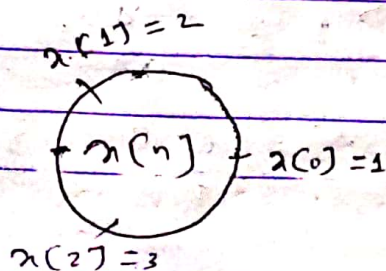


$$y[m] = \{11, 10\}$$

② $x[n] = \{1, 2, 3\}$ & $h[n] = \{4, 5, 6\}$

$y[0], y[1], y[2]$

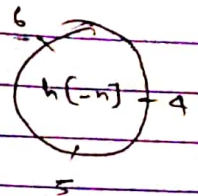
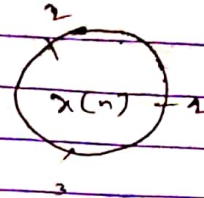
$m = 0, 1, 2.$



$$m=0$$

$$y[0] = \sum_{n=0}^{N-1} x[n]h[n]$$

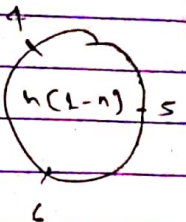
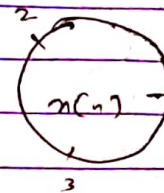
$$y[0] = 1 \times 4 + 2 \times 6 + 3 \times 5 = 31$$



$$m=1$$

$$y[1] = \sum_{n=0}^{N-1} x[n]h[1-n]$$

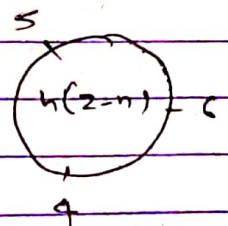
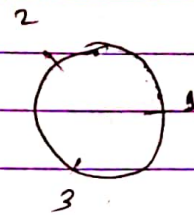
$$y[1] = 1 \times 5 + 2 \times 4 + 3 \times 6 = 31$$



$$m=2$$

$$y[2] = 1 \times 6 + 2 \times 5 + 3 \times 4 = 28$$

$$y[m] = \{31, 31, 28\}$$



checking answer technique.

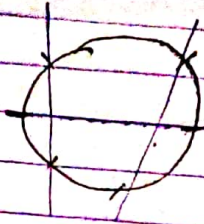
$$x[n] = \{1, 2, 3, 4\}$$

$$h[n] = \{4, 3, 2, 1\}$$

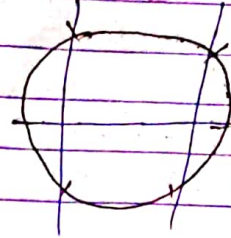
$$\begin{bmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 24 \\ 29 \\ \vdots \end{bmatrix}$$

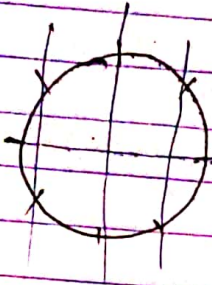
$N = 5$



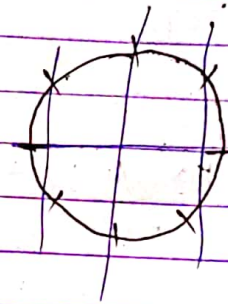
$N = 6$



$N = 7$



$N = 8$



	4	3	2	1
1	4	3	2	1
2	8	6	4	2
3	12	9	6	3
4	16	12	8	4
5	20	15	10	5

{ 4, 11, 20, 30, 40, 26, 24, 13 }

↑

$$x[n] = \{1, 2, 3, 4\} = \{1, 2, 3\} = \{1, 2, 3, 0\}$$

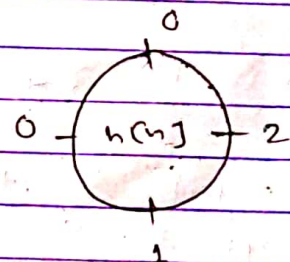
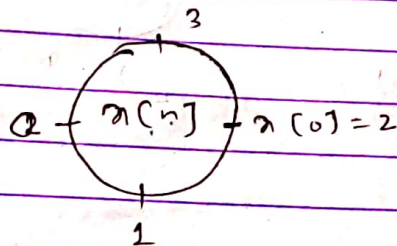
$$h[n] = \{1, 2, 3\} = \{1, 2, 0\} = \{1, 2, 0, 0\}$$

$$m = -1, 0, 1, 2$$

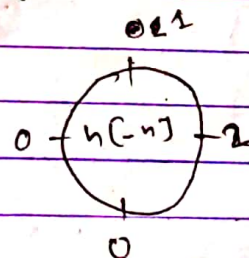
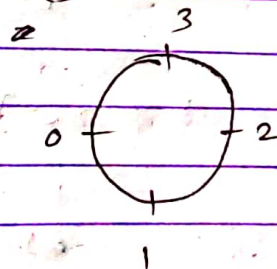
$$y[m] = \sum_{n=0}^{N-1} x[n] h[m-n]$$

for $m=0$

$$y[0] = \sum_{n=0}^{N-1} x[n] h[-n]$$



$$= 2 \times 2 + 1 \times 0 + 3 \times 0 + 0 \times 0$$

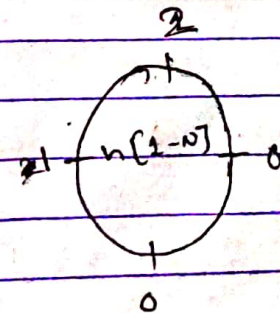
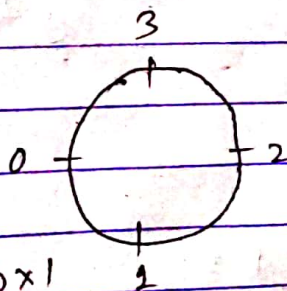


$$= 2 \times 2 + 1 \times 3 + 0 \times 0 + 0 \times 0$$

$$= 7$$

for $m=1$

$$y[1] = 2 \times 0 + 1 \times 3 + 0 \times 2 + 0 \times 0$$

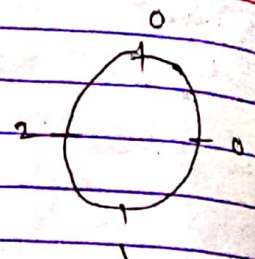
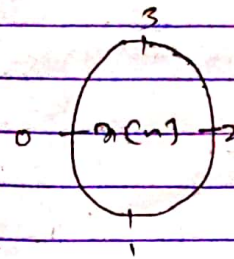


$$= 2 \times 0 + 1 \times 3 + 0 \times 2 + 0 \times 0$$

$$= 6$$

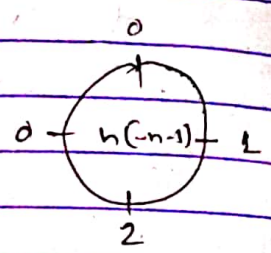
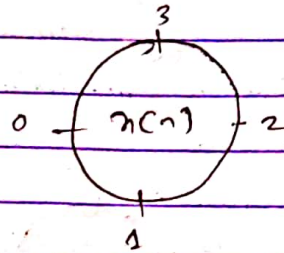
$$m = 2$$

$$2 \times 0 + 3 \times 0 + 2 \times 0 + 1 \times 1 = 1$$



$$m = -1$$

$$2 \times 1 + 3 \times 0 + 0 \times 0 + 1 \times 2 = 4$$



$$\Rightarrow y(m) = \{4, 7, 6, 1\}$$

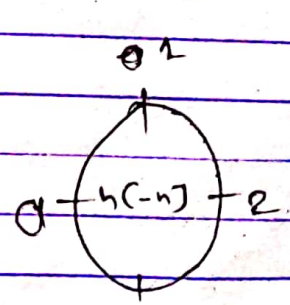
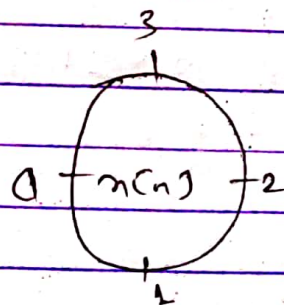
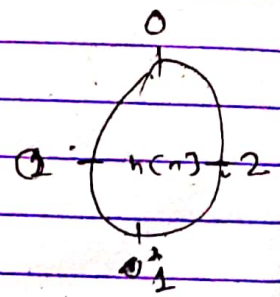
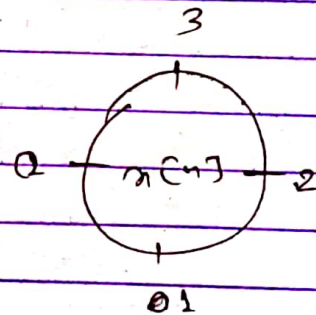
$$\# \quad x(n) = \{0, 1, 2, 3\}$$

$$y(n) = \{0, 1, 2, 3\}$$

$$m = \{-2, -1, 0, 1\}$$

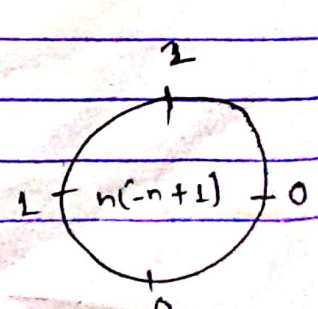
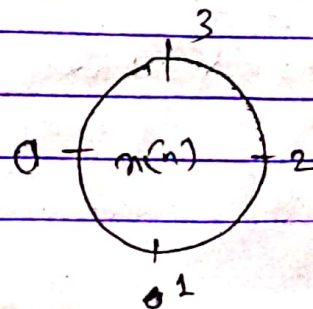
$$\text{for } m = 0$$

$$2 \times 2 + 3 \times 0 + 0 \times 0 + 0 \times 0 = 7$$



$$\text{for } m = 1$$

$$2 \times 0 + 3 \times 2 + 1 \times 0 + 1 \times 0 = 6$$

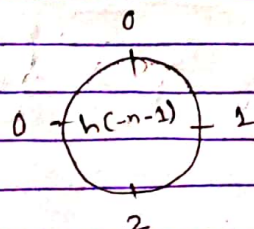
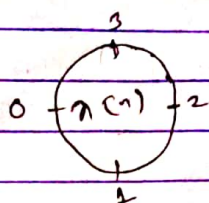


$$m = -1$$

$$1 \times 2 + 3 \times 0 + 0 \times 0 +$$

$$1 \times 2$$

$$\Rightarrow 4$$



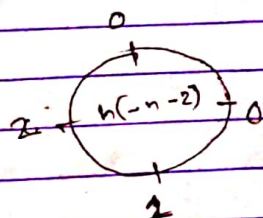
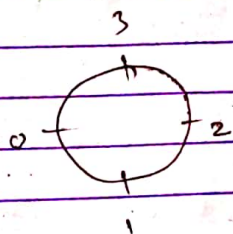
$$m = -2$$

$$\Rightarrow 2 \times 0 + 3 \times 0 + 2 \times 0 +$$

$$1 \times 1$$

$$= 1$$

$$y(n) = \{ \cancel{1} \ 4 \ 7 \ 6 \}$$



Find the circular Convolution of the sequence

$$x(n) = \{1, 2, 3, 4, 5\}$$

$$h(n) = \{4, 3, 2, 1\}$$

find the 5 point circular Convolution

$$x(n) = \{1, 2, 3, 4, 5\}$$

$$h(n) = \{4, 3, 2, 1\}$$

find the circular convolution of the sequence

$$x(n) = \{1, 2, 3, 4, 5\}$$

$$h[n] = \{4, 3, 2, 1\} \Rightarrow \{0, 4, 3, 2, 1\}$$

Circular Convolution

Property $y[n] = x[n] \otimes h[n] \xleftrightarrow[N\text{-point DFT}]{} Y[k] = x[k] \cdot h[k]$

multiplication property. $= \sum_{n=0}^{N-1} x[n] h[m-n]_N$

multiplication property. $y[n] = x[n] \cdot h[n] \xleftrightarrow[N.P. \text{ DFT}]{} Y[k] = \frac{1}{2\pi} [x[k] \otimes h[k]]$

Efficient Computation of DFT (FFT):-

we know that,

$$X(k) = \sum_{n=0}^{N-1} x(n) w_N^{kn} \quad \text{where } k = 0, 1, \dots, N-1$$

This is the eqⁿ for finding DFT (Direct computation method). There are N^2 times complex multiplication and $N(N-1)$ complex addition in DFT. i.e. it will take very large time for processing the output. Hence, the process seems to be slow. Thus, in order to reduce computational complexity and processing time FFT algorithm is used. Mathematically, for DFT calculation $N = r^v$ where r = radix or base of FFT algorithm, v = no. of stages in FFT algorithm.

In our case, $r = 2$ so it is called radix-2 algorithm.

$$N = r^v$$

For,

$$N = 2, \quad v = 1$$

$$N = 4, \quad v = 2$$

$$N = 8, \quad v = 3$$

As, $N = 2^v$

$$\therefore \boxed{v = \log_2 N}$$

For FFT algorithm,

$$\text{Total no. of addition} = N \log_2 N$$

$$\text{Total no. of multiplication} = \frac{N}{2} \log_2 N$$

How fast is FFT compare to DFT?
Speed Improvement Factor (SIF) = $\frac{\text{Total no. of multiplication by direct computation}}{\text{Total no. of multiplication using FFT}}$

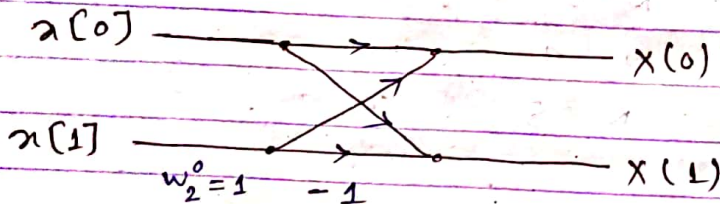
S.No	N	Direct Computation	FFT $\frac{N}{2} \log_2 N$	SIF
1	2	4	1	1
2	4	16	4	4
3	8	64	12	53

Types of FFT algo

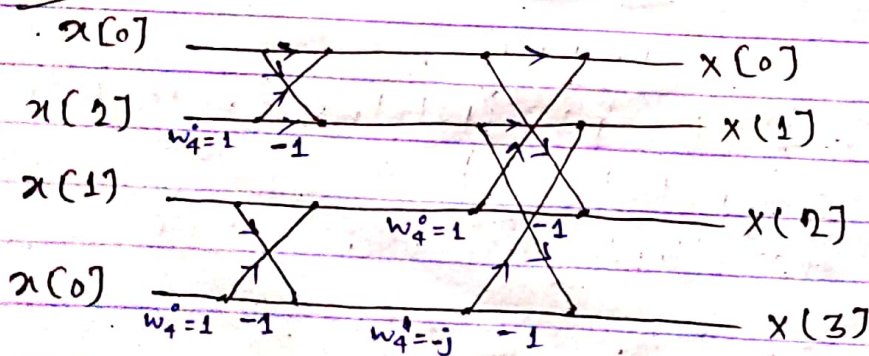
- (1) Decimation In Time - FFT algorithm (DIT-FFT)
- (2) Decimation In Frequency - FFT algorithm (DIF-FFT)
↳ Do yourself.

DIT-FFT

N=2



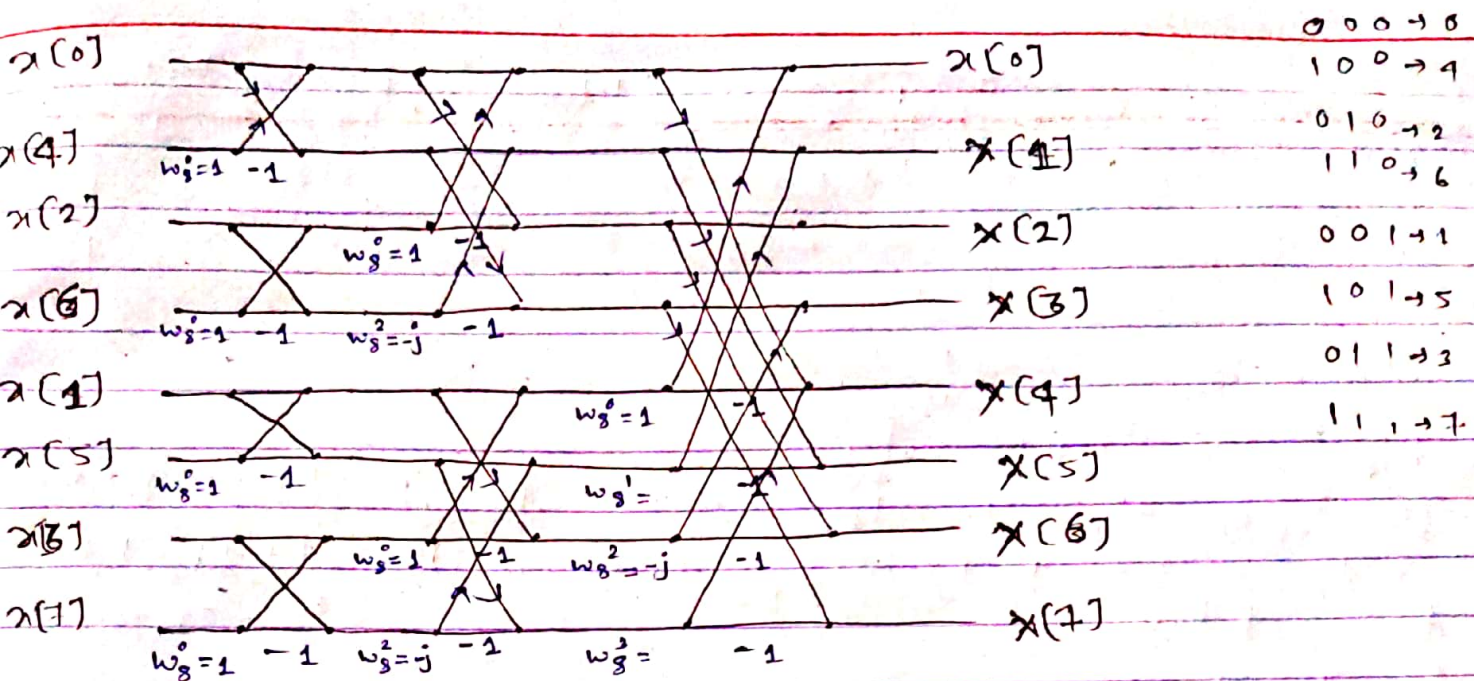
N=4



Take the
binary
from
reverse

0	0	→ 0
1	0	→ 2
0	1	→ 1
1	1	→ 3

$N=8$



Formulas.

$$\left. \begin{aligned} w_8^1 &= 0.707 - j0.707 \\ w_8^3 &= -0.707 - j0.707 \end{aligned} \right\}$$

$x(n) = \begin{cases} 1, & 0 \leq n \leq 5 \\ 0, & \text{elsewhere.} \end{cases}$

$$\begin{aligned} X(0) &= x(0) + x(4) + x(2) + x(6) + x(1) + x(5) + x(3) + x(7) \\ X(4) &= x(0) + x(4) + x(2) + x(6) - x(1) - x(5) - x(3) - x(7) \\ X(2) &= +x(0) + x(4) - x(2) - x(6) + x(1) + x(5) - x(3) - x(7) \end{aligned}$$

$$X(6) = +x(0) + x(4) - x(2) - x(6) + x(1)w_8^2 + x(5)w_8^2 - x(3)w_8^2 - x(7)w_8^2$$

$\therefore X(6)$ is Complex Conjugate of $X(2)$

$$X(1) = +x(0) - x(4) + x(2)w_8^2 - x(6)w_8^2 + x(1)w_8^1 - x(5)w_8^1 + x(3)w_8^2w_8^1 - x(7)w_8^2w_8^1$$

$\therefore X(7)$ is c.c. of $X(1)$

$$X(3) = +x(0) - x(4) - x(2)w_8^2 + x(6)w_8^2 + x(1)w_8^3 - x(5)w_8^3 - x(3)w_8^2w_8^3 + x(7)w_8^2w_8^3$$

$\therefore X(5)$ is c.c. of $X(3)$

Discrete Filter Implementation:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

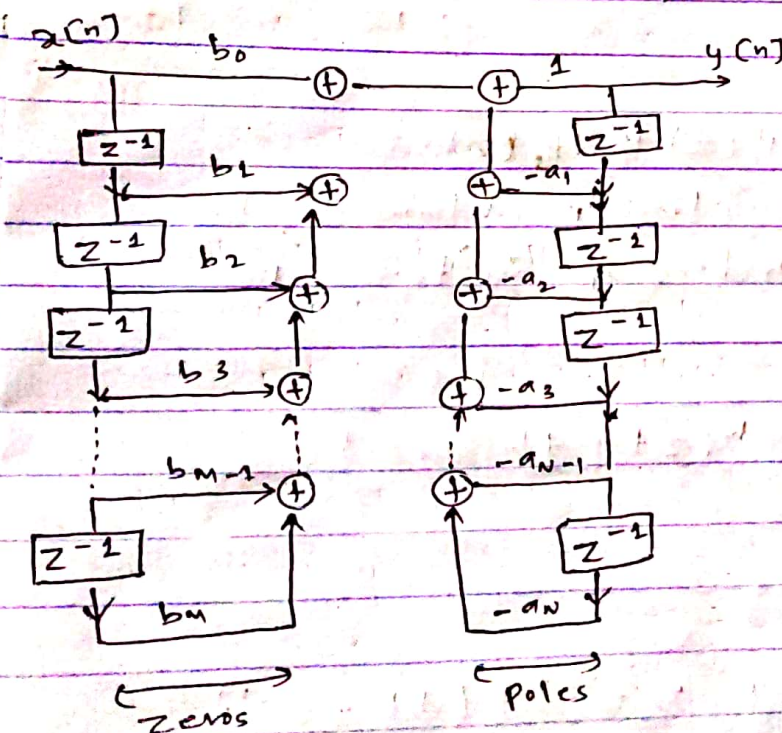
Filter realisation.

(1) Direct form realisation

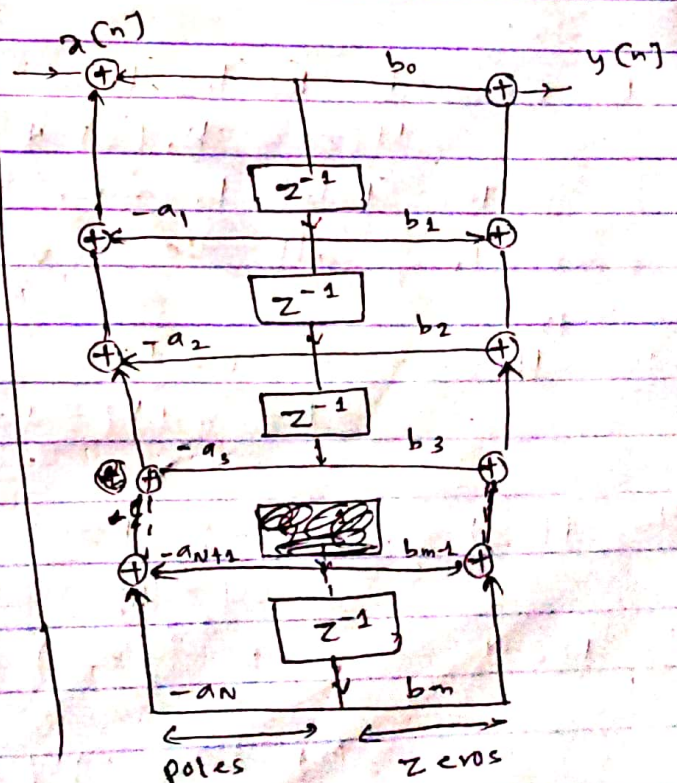
(a) direct Form I }
(b) Direct form II } Summation form

(2) Cascade Realization }
Parallel Realization } Factorized form

(a) Direct Form I



Direct Form II



$$(1) H(z) = \frac{1 + 2z^{-1} + 3z^{-2} + 4z^{-3}}{2 + z^{-1} + z^{-2}}$$

always
should be
1 or 0

$$= \frac{0.5 + z^{-1} + 1.5z^{-2} + 2z^{-3}}{1 + \frac{1}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

Now, form the connection as above.

To obtain numerator and denominator of given transfer function is factored into second order terms such that the transfer function is represented as:

B. Cascade Realisation.

$$H(z) = H_1(z) * H_2(z) * H_3(z) \dots$$

The total transfer function is obtained by connecting all second orders of system in series

$$(1). H(z) = \frac{2(z+2)}{z(z-0.1)(z+0.5)(z+0.4)}$$

$$(2). H(z) = \frac{z^2 - 3z + 2}{z^2 - 7z + 12}$$

$$(3) H(z) = \frac{10 \left(1 - \frac{1}{2}z^{-1}\right) \left(1 - \frac{2}{3}z^{-1}\right) (1 + 2z^{-1})}{\left(1 - \frac{3}{4}z^{-1}\right) \left(1 - \frac{1}{8}z^{-1}\right) \left[1 - \left(\frac{1}{2} + j\frac{1}{2}\right)z^{-1}\right] \left[1 - \left(\frac{1}{2} - j\frac{1}{2}\right)z^{-1}\right]}$$

$$\text{Zeros, } m = \text{int} \left(\frac{n+1}{2} \right)$$

$$\text{Poles, } n = \text{int} \left(\frac{n+1}{2} \right)$$

$$H(z) = 2 + 2z^{-1} + 2z^{-2}$$

$$1 + 2z^{-1} + 3z^{-2} + 3z^{-3}$$

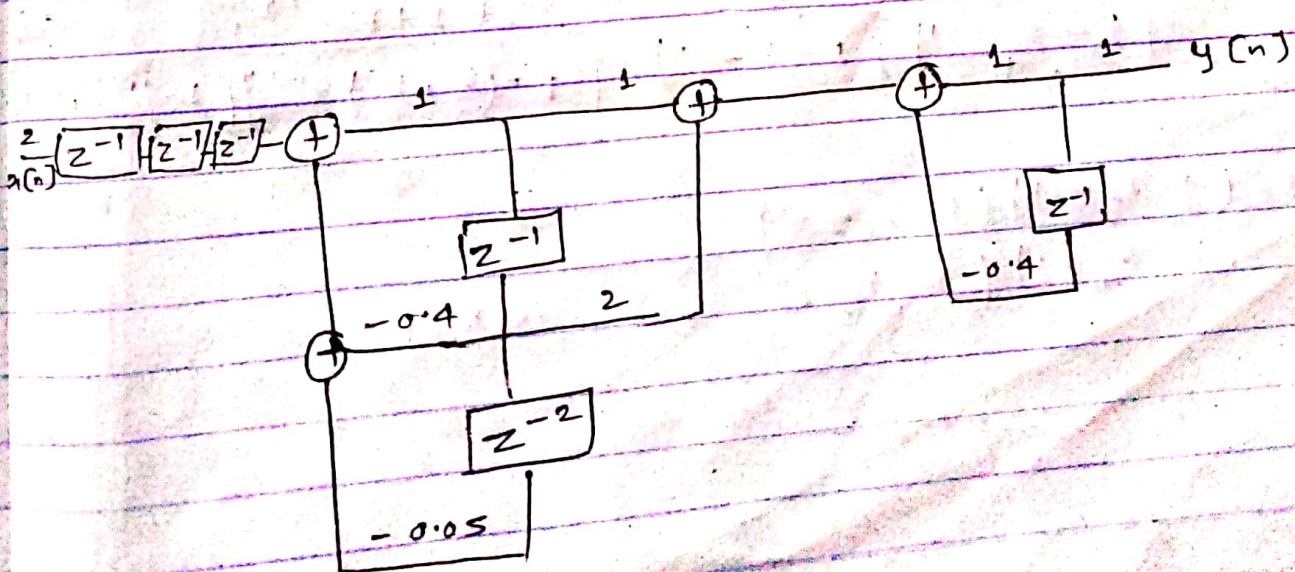
$$H(z) = 2 + 2z^{-1} + 2z^{-3}$$

$$2 + 2z^{-1} + 3z^{-2}$$

$$H(z) = \frac{2z^2 + z + 2}{z^2 + 3z + 3} = \frac{z^2(1 + z^{-1} + 2z^{-2})}{z^2(1 + 3z^{-1} + 3z^{-2})}$$

$$H(z) = \frac{(z-1)(z-2)}{(z-3)(z-4)} = \frac{z^2 - 3z + 2}{2z^2 - 7z + 12}$$

$$\begin{aligned} & \underline{(1)} \quad \frac{2z^{-1}(1+2z^{-1})}{z^4(1-0.1z^{-1})(1+0.5z^{-1})(1+0.4z^{-1})} \\ & = \frac{2z^3(1+2z^{-1})}{(1-0.1z^{-1})(1+0.5z^{-1})(1+0.4z^{-1})} \\ & = 2z^{-3} * \frac{(1+2z^{-1})}{(1-0.1z^{-1})(1+0.5z^{-1})} * \frac{1}{(1+0.4z^{-1})} \\ & = 2z^{-3} * \frac{1+2z^{-1}}{1+0.4z^{-1}-0.05z^{-2}} * \frac{1}{1+0.4z^{-1}} \end{aligned}$$



$$\begin{aligned}
 (2) \quad H(z) &= \frac{z^2 - 3z + 2}{z^2 - 7z + 12} \\
 &= \frac{z^2(1 - 3z^{-1} + 2z^{-2})}{z^2(1 - 7z^{-1} + 12z^{-2})} \cdot \frac{(z-1)(z-2)}{(z-3)(z-4)} \\
 &= \frac{1 - 3z^{-1} + 2z^{-2}}{1 - 7z^{-1} + 12z^{-2}} \\
 &= \frac{(1 - z^{-1})(1 - 2z^{-1})}{(1 - 3z^{-1})(1 - 4z^{-1})} \\
 &= \frac{1 - z^{-1}}{1 - 3z^{-1}} * \frac{1 - 2z^{-1}}{1 - 4z^{-1}}
 \end{aligned}$$

$$\text{Zeros } m = \text{int} \left(\frac{M+1}{2} \right)$$

$$= \frac{\cancel{2}+1}{2} = \text{int} \left(\frac{1.5}{1} \right) = 1$$

$$\text{Poles } n = \text{int} \left(\frac{N+1}{2} \right)$$

$$= \text{int} \left(\frac{2+1}{2} \right) = \text{int} \left(\frac{1.5}{1} \right) = 1$$

$$\begin{aligned}
 (3) \quad H(z) &= \frac{10 (1 - \frac{1}{2}z^{-1}) (1 - \frac{3}{4}z^{-1}) (1 + 2z^{-1})}{(1 - \frac{3}{4}z^{-1}) (1 - \frac{1}{8}z^{-1}) \left[1 - \left(\frac{1}{2}j\frac{1}{2} \right) z^{-1} \right] \left[1 - \left(\frac{1}{2}-j\frac{1}{2} \right) z^{-1} \right]} \\
 &= \frac{10 (1 - \frac{1}{2}z^{-1}) (1 - \frac{3}{4}z^{-1})}{(1 - \frac{3}{4}z^{-1}) (1 - \frac{1}{8}z^{-1})} \times \frac{z \times y}{z \times y}
 \end{aligned}$$